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Position Sensorless Direct Torque Control of Brushless DC Motor Drives Based on Sliding Mode Observer Using NSGA-II Algorithm Optimization

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Abstract—This paper presents a direct torque control technique for Brushless DC motors with non-sinusoidal back electromotive force. Direct torque control has some benefits such as faster torque response and reduced torque ripple for driving the Brushless DC motors. The Sliding mode observer, which is robust to parameter uncertainties, can be used to estimate the back-EMF and generate the torque. It is employed to estimate the non-sinusoidal back-EMF waveform in a Brushless DC motor using only the measurements of the stator currents. A modified version of Non-Dominated Sorting Genetic Algorithm (NSGA-II) is used as an effective optimization tool for regulating the PID controller parameters of speed controller and selection of four gains of sliding mode observer. The relation between PID controller and the sliding mode is also satisfied using NSGA-II. The effectiveness of proposed system has been validated by simulation results.

Keywords- BLDC; DTC; sliding mode observer; non-dominated sorting genetic algorithm (NSGA-II)

I. INTRODUCTION

BLDC motors have come to dominate many applications such as home appliances, automobiles, information technology equipment, industries, transportation, aerospace, defense equipment, power tools, sound equipment, and laboratory medical equipment in wide range of power. Due to better speed versus torque characteristics, high dynamic response, high efficiency, long operating life, noiseless operation and higher speed ranges, BLDC motors excel brushed DC and induction motors.

Direct Torque Control (DTC) method proposed in [1] and [2] has been utilized to drive the BLDC motors. DTC, which is also known as conventional DTC, has the features like fast torque response, simple and robust design.

For a three-phase BLDC motor, only six discrete rotor positions per electrical cycle are needed to feed a rectangular current in phase with the back-EMF. For optimum performance, commutation has to be carried out at precise rotor positions. Hall sensors are normally used for this purpose. However, they introduce a number of drawbacks; they increase

the cost and need special mechanical arrangements to be mounted. Furthermore, Hall sensors are temperature-sensitive. They also deteriorate the system reliability by adding up extra components and wiring. Consequently, there has been significant effort on practical implementation of sensorless techniques to spot commutation times. These methods are based on a variety of strategies, including terminal voltage sensing, back-EMF integration, third harmonic of the back-EMF sensing, detection of freewheeling diode current, flux calculation based methods and observer based methods like Kalman filter and sliding mode observer.

Sliding mode observer is used for linear and nonlinear systems. This type of observer is very useful and was developed for many reasons such as, working with reduced observation error dynamics, possibility of obtaining a step-by-step design, finite time convergence for all the observables states, designing an observer for non-smooth systems under some conditions, and robustness under parameter variations is possible [3]-[7].

This paper proposes a sliding-mode observer to estimate the trapezoidal back-EMF of a BLDC motor. The estimated back-EMF is in turn used to achieve estimated torque, rotor position and speed of the motor using only the measurements of the stator currents. To achieve the optimal response, the optimal PID controller parameters and sliding mode gains must be applied. It should be solved as a multiobjective problem (MOP). A modified version of Non-Dominated Sorting Genetic Algorithm, NSGA-II, was developed to solve MOP problem [8]. Effectiveness of proposed method can be deduced from simulation results.

II. BLDC MOTOR OPERATION PRINCIPLE

BLDC motors are truly an inside-out DC commutator with the mechanical commutator replaced by an electronic switching converter.

The stator generates the magnetic field to make the rotor rotate. Hall Effect sensors detect the rotor position as the commutating signals. Therefore, BLDC motors use permanent

magnets instead of coils in the armature and do not need brushes. For the three phase BLDC motors the back-EMF and phase current waveforms with 120° conduction mode are shown in Fig.1, [9], [10].

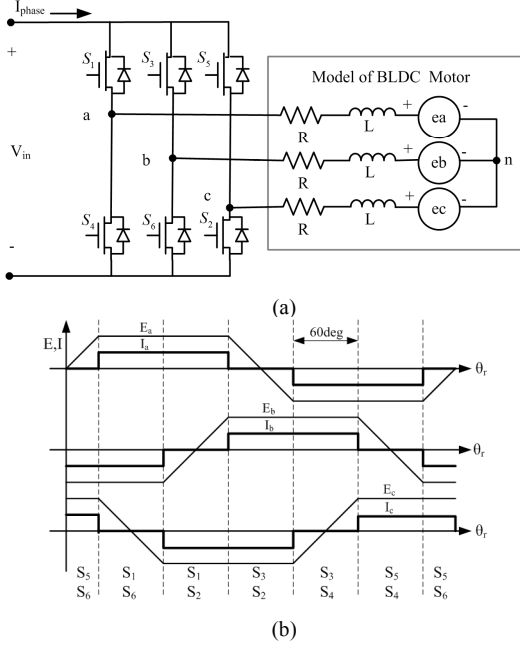


Figure 1. (a) Configuration of BLDC motor drive system, (b) back EMF pattern and reference current generation

The analysis of a BLDC motor is represented as the following equations:

$$\begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L-M & 0 & 0 \\ 0 & L-M & 0 \\ 0 & 0 & L-M \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (1)$$

where u_a , u_b and u_c are the phase voltages, i_a , i_b and i_c are the phase currents, e_a , e_b and e_c are the phase back-EMF waveforms, R is the phase resistance, L is the self inductance of each phase and M is the mutual inductance between any two phases.

So the electromagnetic torque can be obtained as:

$$T_e = \frac{1}{\omega_r} (e_a i_a + e_b i_b + e_c i_c) \quad (2)$$

where ω_r is the mechanical speed of the rotor.

The dynamic equation is:

$$\frac{d}{dt} \omega_r = \frac{1}{J} (T_e + T_L + B \omega_r) \quad (3)$$

where B is the damping constant, J is the moment of inertia of the drive and T_L is mechanical torque.

III. DIRECT TORQUE CONTROL OF BLDC MOTOR DRIVE USING TWO – PHASE CONDUCTION MODE

DTC was originally developed for induction machine drives, and controls the flux linkage and electromagnetic torque directly. The key issue in the DTC of a BLDC motor drive in the constant torque region is to estimate the electromagnetic torque correctly. Equation (4) which is given in the stationary reference frame can be used for the electromagnetic torque calculation [1], [2].

$$\begin{aligned} T_e &= \frac{3}{2} \frac{P}{2} \frac{1}{\omega_e} (e_{sq} i_{sq} + e_{sd} i_{sd}) \\ &= \frac{3}{2} \frac{P}{2} [k_q(\theta_e) i_{sq} + k_d(\theta_e) i_{sd}] \end{aligned} \quad (4)$$

where θ_e is the electrical rotor position, ω_e is the electrical rotor speed and $k_q(\theta_e)$, $k_d(\theta_e)$, e_{sq} , e_{sd} and i_{sq} , i_{sd} are the stationary reference frame (dq-axes) back-EMF constants, motor back-EMFs and stator currents, respectively.

Since the right side of (4) does not involve the rotor speed in the denominator there will be no problem in the estimating the torque at zero and near zero speeds. Therefore, it is used in the proposed control system. Fig. 2 shows the basic functional blocks used to implement the core of the DTC scheme. The torque comparator contains a hysteresis control block to compare the torque reference with its corresponding actual value. The actual speed is compared to the speed command to form the torque command from the PI speed regulator. Although in conventional DTC, torque and flux are considered in the overall control system but three reasons are given for eliminating the flux control. First, since the magnitude of the phase back-EMF is below 50% of the dc-link voltage in the constant torque region, there is no need to control the flux amplitude. Second, with the two-phase conduction mode, sudden sharp dips in the stator flux linkage locus occur that complicate the control scheme. The size of these sharp dips is unpredictable. Third, regardless of the stator flux linkage amplitude, the phase currents tend to match with the flat top portion of the corresponding trapezoidal back-EMF to generate a constant torque.

As it can be seen from Table I, the switching pattern of the inverter can be determined according to the torque status from the output of hysteresis. Therefore, in the DTC of a BLDC motor drive the flux error F_{st} in the voltage vector selection look-up table is always selected as zero and only the torque error T_{st} is used depending on the error level of the actual torque from the reference torque. If the reference torque is greater than the actual torque, within the hysteresis bandwidth, the torque error T_{st} is defined as "TI", otherwise it will be defined as "TD", as shown in Table I. By means of this method BLDC motor will be successfully driven.

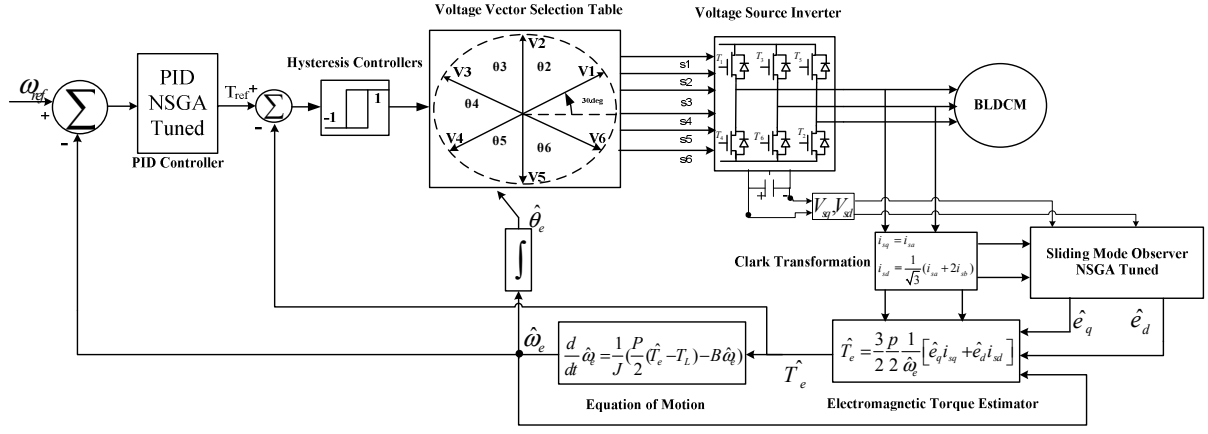


Figure 2. Overall block diagram of the DTC of a BLDC motor drive

TABLE I. SWITCHING STATE SELECTOR FOR DTC, TORQUE INCREASE (TI), TORQUE DECREASE (TD) AND NO-CHANGE IN FLUX (F)

F_{ST}	T_{ST}	SECTOR					
		Θ_1	Θ_2	Θ_3	Θ_4	Θ_5	Θ_6
F	TI	$V_2(001001)$	$V_3(011000)$	$V_4(010010)$	$V_5(000110)$	$V_6(100100)$	$V_1(100001)$
	TD	$V_5(000110)$	$V_6(100100)$	$V_1(100001)$	$V_2(001001)$	$V_3(011000)$	$V_4(010010)$

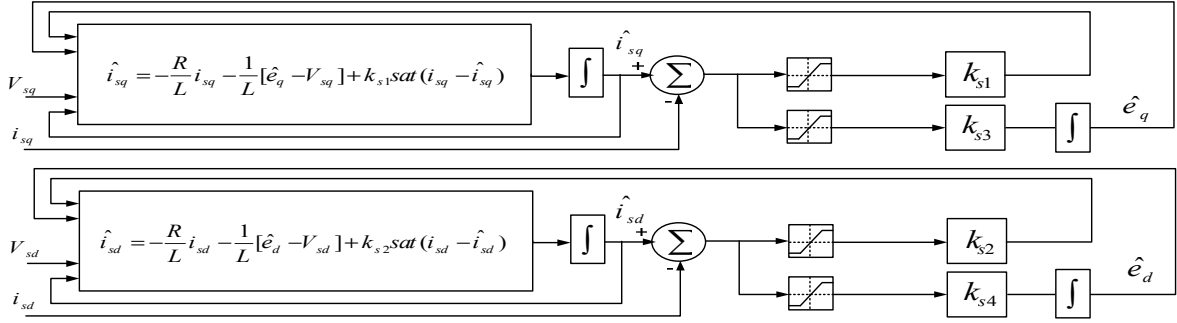


Figure 3. Sliding mode observer block diagram

IV. SLIDING MODE OBSERVER DESIGN

Sliding mode observers have unique properties, in that the ability to generate a sliding motion on the error between the measured plant output and the output of the observer. This ability ensures that a sliding mode observer produces a set of state estimates that are precisely commensurate with the actual output of the plant. The BLDC motor equations are presented in (5):

$$\begin{aligned}
 \frac{di_{sq}}{dt} &= -\frac{R}{L} i_{sq} - \frac{1}{L} e_{sq} + \frac{1}{L} u_{sq} \\
 \frac{di_{sd}}{dt} &= -\frac{R}{L} i_{sd} - \frac{1}{L} e_{sd} + \frac{1}{L} u_{sd} \\
 \frac{de_{sq}}{dt} &= 0 \\
 \frac{de_{sd}}{dt} &= 0
 \end{aligned} \quad (5)$$

If the sampling period is significantly less than the electrical and mechanical time constants then the back-EMF value can be assumed to be constant during each sampling period so, $de_{sq}/dt = 0$, $de_{sd}/dt = 0$ [5].

The theory of sliding mode is represented in [11]. To decrease the chattering effects near the sliding mode surface, a sign function is replaced with the saturation function. Therefore, the sliding mode observer is proposed as:

$$\begin{aligned}
 \dot{\hat{i}}_{sq} &= -\frac{R}{L} i_{sq} + \frac{1}{L} (-\hat{e}_{sq} + u_{sq}) + k_{s1} sat(i_{sq} - \hat{i}_{sq}) \\
 \dot{\hat{i}}_{sd} &= -\frac{R}{L} i_{sd} + \frac{1}{L} (-\hat{e}_{sd} + u_{sd}) + k_{s2} sat(i_{sd} - \hat{i}_{sd}) \\
 \dot{\hat{e}}_{sq} &= k_{s3} sat(i_{sq} - \hat{i}_{sq}) \\
 \dot{\hat{e}}_{sd} &= k_{s4} sat(i_{sd} - \hat{i}_{sd})
 \end{aligned} \quad (6)$$

By setting $e = x - \hat{x}$, error dynamics are represented in (7):

$$\begin{aligned}
\dot{e}_1 &= -\frac{1}{L}e_3 + k_{s1}sat(e_1) \\
\dot{e}_2 &= -\frac{1}{L}e_4 + k_{s2}sat(e_2) \\
\dot{e}_3 &= -k_{s3}sat(e_1) \\
\dot{e}_4 &= -k_{s4}sat(e_2)
\end{aligned} \tag{7}$$

Positive definite Lyapunov function is defined as sliding surface in (8):

$$S(e) = \frac{1}{2}(e_1^2 + e_2^2) \tag{8}$$

The time derivative of $S(e)$ is shown in (9):

$$\begin{aligned}
\dot{S}(e) &= e_1\dot{e}_1 + e_2\dot{e}_2 \\
&= e_1[-\frac{1}{L}e_3 + k_{s1}sat(e_1)] + e_2[-\frac{1}{L}e_4 + k_{s2}sat(e_2)]
\end{aligned} \tag{9}$$

The observer gains must be chosen so that the $S \cdot \dot{S} < 0$ limitation is satisfied. The NSGA-II is applied to tune the observer gains. A block diagram of the sliding mode observer is given in Fig. 3.

V. MULTIOBJECTIVE OPTIMIZATION PROBLEM

Multiojective optimization is an appropriate tool for handling different incommensurable objectives with conflicting/supporting relations, or not having any mathematical relation with each other.

A. Objective Functions and Constraints

This problem has two objective functions which must be simultaneously minimized. These objective functions are defined for tuning the PID controller parameters and the sliding mode observer gains.

The transfer function of a PID controller is given by:

$$G(s) = k_p + \frac{k_i}{s} + k_d s \tag{10}$$

where k_p , k_i and k_d are the proportional, integral and derivative gains, respectively.

The system response can be evaluated by the overshoot (M_p), rise time (t_r), settling time (t_s), and steady-state error (E_{ss}), [12]. To improve the system performance by considering these parameters, the first objective function is defined as:

$$f_1(k_p, k_i, k_d) = (1 - e^{-\beta})(M_p + E_{ss}) + e^{-\beta}(t_s - t_r) \tag{11}$$

where β is the weight factor.

Decreasing the rise time and settling time can be achieved by using a weight factor less than 0.7; increasing this parameter to more than 0.7, will result in the reduction of overshoot and steady state error.

According to (8), the second objective function can be defined as:

$$f_2(k_{s1}, k_{s2}, k_{s3}, k_{s4}) = \frac{1}{2}(e_1^2 + e_2^2) \tag{12}$$

Since $S \cdot \dot{S} < 0$ must be satisfied and according to (8) and (9), some constraints are extracted as follows:

if $e_1 > 0$, the switching gain, $k_{s1} > -\frac{1}{L}e_3$

if $e_1 < 0$, the switching gain, $k_{s1} > \frac{1}{L}e_3$

if $e_2 > 0$, the switching gain, $k_{s2} > -\frac{1}{L}e_4$

if $e_2 < 0$, the switching gain, $k_{s2} > \frac{1}{L}e_4$

B. Solution Method

NSGA-II is one of the most efficient multiobjective evolutionary algorithms using elitist approach which has great advantages in comparison to other evolutionary algorithms. The general trend of NSGA-II is designed to find a set of the best PID controller gains and sliding mode observer parameters is as; initially, a random parent population is created, each solution is assigned as fitness (or rank) equal to its non-domination level (1 is the best level, 2 is the next-best level and so on). Thus the minimization of fitness is assumed. In addition to the fitness value, a parameter called "crowding distance" is also calculated for each individual. The crowding distance is a measure of how close an individual is to its neighbors. Parents are selected from the population by using binary tournament selection based on the rank and crowding distance. The selected population generates offsprings from crossover and mutation operators. The current population and current offsprings is sorted again based on non-domination and the best N individuals are selected, where N is the population size. The selection is based on the rank and crowding distance in the last solution [8], [13].

Table II shows the optimal sliding mode and PID controller parameters that are obtained by NSGA-II. Also, the parameters of NSGA-II are given in this table.

TABLE II. NSGA-II, OPTIMAL SLIDING MODE OBSERVER AND PID PARAMETERS

Parameter	Value
Total Number of Generations, Gen	50
Population Size, N	20
Proportional Gain, k_p	73.3254
Integrator Gain, k_i	0.51326
Derivative Gain, k_d	0.30096
k_{s1}	202.4226
k_{s2}	401.0967
k_{s3}	-25.2238
k_{s4}	-31.0089

VI. SIMULATION RESULTS

In order to evaluate the effectiveness of the proposed method, the drive system shown in Fig. 2 was simulated by MATLAB/Simulink. Motor parameters used for the simulation are given in Table III.

TABLE III. BLDC MOTOR PARAMETER USED FOR SIMULATION

Parameter	Value
Number of Poles	2
DC Link Voltage	300 V
Rated Speed	1500 rpm
Phase Resistance	0.4 Ω
Self Inductance	13 mH
Load Torque	3 N.m
Moment of Inertia	0.004 kg.m ²

The actual and the estimated d-q frame back-EMFs are shown in Figs. 4 and 5. Figs. 4(a) and 5(a) are achieved from back-EMF shaping and Figs. 4(b) and 5(b) are estimated by sliding mode observer. It is obvious shown that the estimated back-EMFs are accurate and the sliding mode observer is a good option for back-EMFs estimation without requiring look up table and position sensors.

Actual and estimated electrical rotor speeds are shown in Fig. 6. The calculation trend of estimated electrical rotor speed is given in Appendix. By comparing Fig. 6(a) and Fig. 6(b), a good conformity of the actual and estimated speed can be observed. Furthermore, speed rises steadily from stand still and reaches the steady state without any serious distortion.

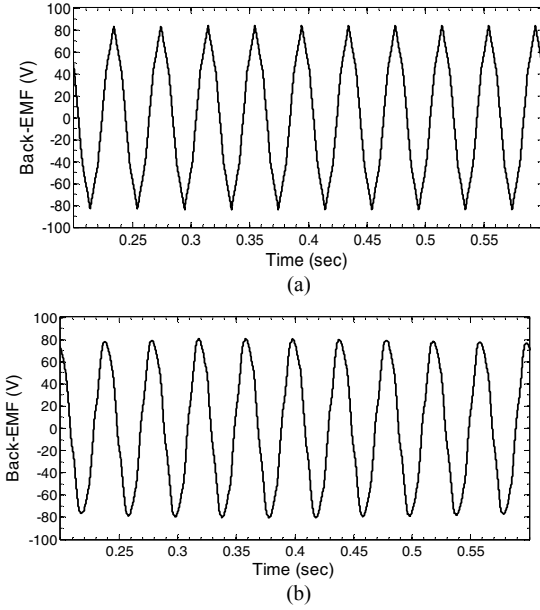


Figure 4. Back-EMF (e_{sq}), (a) Actual, (b) Estimated

The stator phase currents are depicted in Fig. 7. Despite the elimination of position sensors, the waveforms of the phase currents are appropriate. Fig. 8 shows the estimated electromagnetic torque that is calculated according to the Appendix.

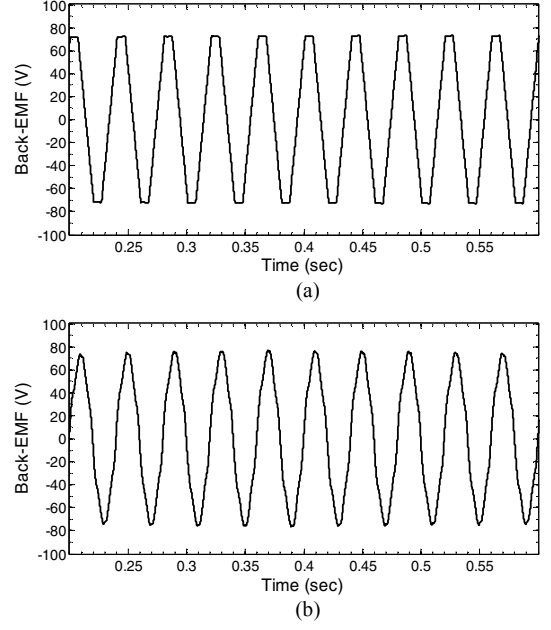


Figure 5. Back-EMF (e_{sd}), (a) Actual, (b) Estimated

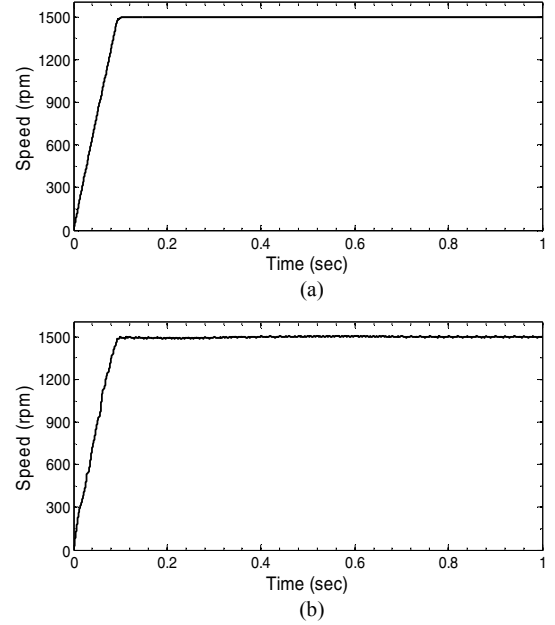


Figure 6. Rotor speed, (a) Actual, (b) Estimated

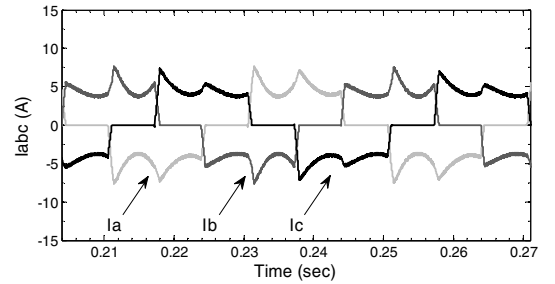


Figure 7. Steady state motor phase currents

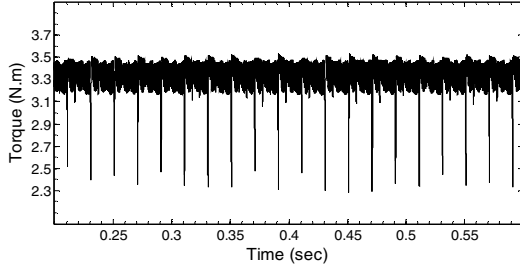


Figure 8. Estimated electromagnetic torque

VII. CONCLUSION

In this paper, a sliding mode observer design was proposed for estimating the trapezoidal back-EMF of a BLDC motor. DTC offers some advantages such as; simple algorithm, simplicity of implementation, faster torque response, reduced torque ripple and less sensitivity to the variations of parameters, so the proposed system has utilized DTC method in order to benefit from the mentioned advantages. The effectiveness of the proposed sensorless method depends on proper selection of sliding mode observer and PID controller parameters. Therefore NSGA-II algorithm was applied to tune these parameters by considering the relation between the PID controller and the sliding mode observer. The effectiveness of the proposed sensorless method has been verified through simulation and the results were presented.

APPENDIX

Consider the back-EMFs that are estimated in section IV, the torque, electrical rotor speed, and position can be calculated.

According to the estimated back-EMF and measured stator currents, the estimated torque is calculated as follows:

$$\hat{T}_e = \frac{3}{2} \frac{P}{2} \frac{1}{\hat{\omega}_e} (\hat{e}_q i_{sq} + \hat{e}_d i_{sd}) \quad (13)$$

where $\hat{e}_q$ and $\hat{e}_d$ are the estimated d-q frame back-EMFs, $\hat{\omega}_e$ is the estimated electrical rotor speed.

Estimated electrical rotor speed and subsequently electrical position of the rotor are calculated as follow:

$$\frac{d}{dt} \hat{\omega}_e = \frac{1}{J} \left[\frac{P}{2} (\hat{T}_e + T_L) + B \hat{\omega}_e \right] \quad (14)$$

$$\hat{\omega}_e = \frac{d}{dt} \hat{\theta}_e \quad (15)$$

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